Bob chooses to measure the horizontal spin instead, he will get spin-right and spin-left with fifty-fifty probability. We can say that if Bob measures the vertical spin, the correlation between his result and Alice's will be 100 percent (we know exactly what he'll get), whereas if he measures horizontal spin, there will be 0 percent correlation (we have no idea what he will get).

So what if Bob, growing bored all by himself in a spaceship orbiting Alpha Centauri, decides to measure the spin of his particle along some axis in between the horizontal and vertical? (For convenience imagine that Alice and Bob actually share a large number of entangled Bell pairs, so they can keep doing these measurements over and over, and we only care about what happens when Alice observes spin-up.) Then Bob will usually, but not always, observe the spin to be pointed along whatever direction is more closely aligned with the vertical "up." In fact, we can do the maths: if Bob's axis is at 45 degrees, exactly halfway between vertical and horizontal, there will be a 71 percent correlation between his results and Alice's. (That's one over the square root of two, if you're wondering where the number comes from.)

What Bell showed, under certain superficially reasonable assumptions, is that this quantum-mechanical prediction is impossible to reproduce in any local theory. In fact, he proved a strict inequality: the best you can possibly do without some kind of spooky action at a distance would be to achieve a 50 percent correlation between Alice and Bob if their measurements were rotated by 45 degrees. The quantum prediction of 71 percent correlation violates Bell's inequality. There is a distinct, undeniable difference between the dream of simple underlying local dynamics, and the real-world predictions of quantum mechanics.

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I presume you are currently thinking to yourself, "Hey, what do you mean that Bell made superficially reasonable assumptions? Spell them out. I'll decide for myself what I find reasonable and what I don't."

Fair enough. There are two assumptions behind Bell's theorem in particular that one might want to doubt. One is contained in the simple idea that Bob "decides" to measure the spin of his qubit along a certain axis. An element of human choice, or free will, seems to have crept into our theorem